

GROWTH INTERN ASSIGNMENT

Jar — Growth Intern Assignment

Sales analysis · App exploration · Product exploration

[Candidate name] · 9 October 2026

Analysis in Python · all figures and numbers regenerated by `python main.py`

Contents

Executive summary	3
1. Data and method	4
1.1 Join validation	4
1.2 Date repairs	4
1.3 Definitions	4
2. Q1 Part 1 — Sales and profitability	5
2.1 Category scorecard	5
2.2 Why the categories differ	6
3. Q1 Part 2 — Target achievement (Furniture)	9
3.1 Month-over-month change in target	9
3.2 Target vs actual: where expectations and reality diverge	10
3.3 Aligning targets with performance	10
4. Q1 Part 3 — Regional performance	12
4.1 Top 5 states by order count	12
4.2 Regional disparities	13
4.3 Where to focus	14
5. Q2 — App exploration	15
5.1 Five things that work well	15
5.2 Five areas to improve	15
6. Q3 — Product exploration	17
6.1 Starting point: Jar's real strengths, and one correction	17
6.2 Opportunity cards	18
6.3 Sequenced roadmap	19
6.4 Sources (Sections 5–6)	19
Appendix A — Assumptions log	21
Appendix B — Detail tables	22
Appendix C — Reproducibility	25
Appendix D — Q2 hands-on checklist (≈ 20 minutes)	26

Executive summary

Electronics leads the business on scale, Clothing earns the most profit, and Furniture barely breaks even.

Electronics books the most sales (₹1,65,267, 38.3% of the total) and the highest profit per order (₹51.44).

Clothing has the highest total profit (₹11,163, 46.6% of company profit) and the best margin (8.03%). Furniture brings in 29.5% of sales but only 9.6% of profit, at a 1.81% margin.

The losses point to discounting, not volume, and they are concentrated. Two sub-categories lose money outright: Tables (-₹4,011) and Electronic Games (-₹1,236). Without Tables, Furniture's margin rises from 1.81% to 6.03%. In 16 of 17 sub-categories, loss-making lines sell at a lower unit price than profitable lines (median 0.55×). Median quantity is the same on both (3 units), which points to price erosion rather than order size.

The Furniture target is phased for a flat year; demand is back-loaded. The target rises smoothly from ₹10,400 to ₹11,800, and no month moves more than 1.89%. Actual sales missed target for 7 straight months (Apr–Oct, 64.2% attainment), then beat it in 4 of the last 5. The annual target was nearly met (95.7%). Re-phasing the same annual total to company seasonality cuts the average monthly error (MAPE) from 54.4% to 19.5%.

Regional growth should focus on margin repair in large cities and expansion in high-margin small states. Madhya Pradesh and Maharashtra hold 38.2% of orders. Mumbai brings in 64.9% of Maharashtra's sales at a 2.65% margin, against 13.56% in Pune. 4 states lose money (Tamil Nadu, Punjab, Andhra Pradesh and Bihar; -₹3,642 combined). Uttar Pradesh, Delhi, West Bengal and Kerala earn 44.2% of profit on 16.8% of sales.

Q2 – App. The evaluation framework rests on Jar's documented features: ₹10 automated saving, vault and insurance trust signals, vernacular UI, liquidity, and spend round-ups. All observations are flagged **[VERIFY]** to be confirmed in a hands-on session; Appendix D has the checklist.

Q3 – Product. The first move is to take Jar's automation onto **regulated** rails. SEBI's November 2025 advisory places digital gold outside its oversight, so Jar's earned trust currently rests on an unregulated product. Recommended sequence: goal-linked Nek savings and a regulated gold SIP now; insured micro-deposits next; sachet insurance after that; gold-backed credit only once compliance is in place.

1. Data and method

1.1 Join validation

Order Details (one row per line item) was **left-joined** to List of Orders on Order ID with pandas `validate="many_to_one"` and an indicator column. Order Details is the left table because it carries the grain of every financial metric. The many-to-one check proves Order ID is unique on the header side, so no line can be duplicated by the join. A hard assertion fails the run if any line is orphaned, instead of letting an inner join drop it silently.

Check	Result
List of Orders rows / unique Order IDs	500 / 500
Order Details rows / unique Order IDs	1,500 / 500
Duplicate Order IDs in List of Orders	0
IDs only in List of Orders / only in Order Details	0 / 0
Rows after merge	1,500 (equals line items)

Every order has 1 to 12 lines. 207 orders span more than one category, which is why "per order" and "per line" averages differ.

1.2 Date repairs

Both date fields were corrupted by spreadsheet auto-conversion. Both were repaired in code, with assertions; the raw files are untouched.

- **Order Date.** 307 cells are `dd-mm-yyyy` text. The other 193 were read by Excel as month-first, so day and month are swapped (for example, 1 April 2018 was stored as 4 January). After swapping them back, dates rise monotonically with the sequential Order ID and span 01 Apr 2018 to 31 Mar 2019 (FY 2018-19). Without the repair, dates are non-monotonic and run into December 2019.
- **Sales Target month.** Labels such as "Apr-18" were stored as *18 April of the current year*. The true month is the stored month, and the true year is 2000 + the stored day. That gives April 2018 to March 2019, exactly overlapping the order period, so actual-vs-target comparisons cover all 12 months.

1.3 Definitions

- **Sales** = sum of `Amount`. **Margin** = $\text{sum(Profit)} \div \text{sum(Amount)}$.
- **Average profit per order (primary)** = a group's total profit $\div$ the number of distinct Order IDs containing at least one of its lines. **Per line item (secondary)** = mean profit across the group's lines. Both definitions rank the three categories identically; the per-order figure is 1.3×–2.4× higher because orders hold several lines.
- **Order count** = distinct Order IDs, never row count.
- **Discount proxy.** The data has no discount field. Realised unit price ($\text{Amount} \div \text{Quantity}$) on loss-making lines is compared with profitable lines *within the same sub-category*.

2. Q1 Part 1 — Sales and profitability

2.1 Category scorecard

Electronics leads on sales and profit per order; Furniture trails on every metric

Category totals across all 500 orders, Apr 2018 - Mar 2019

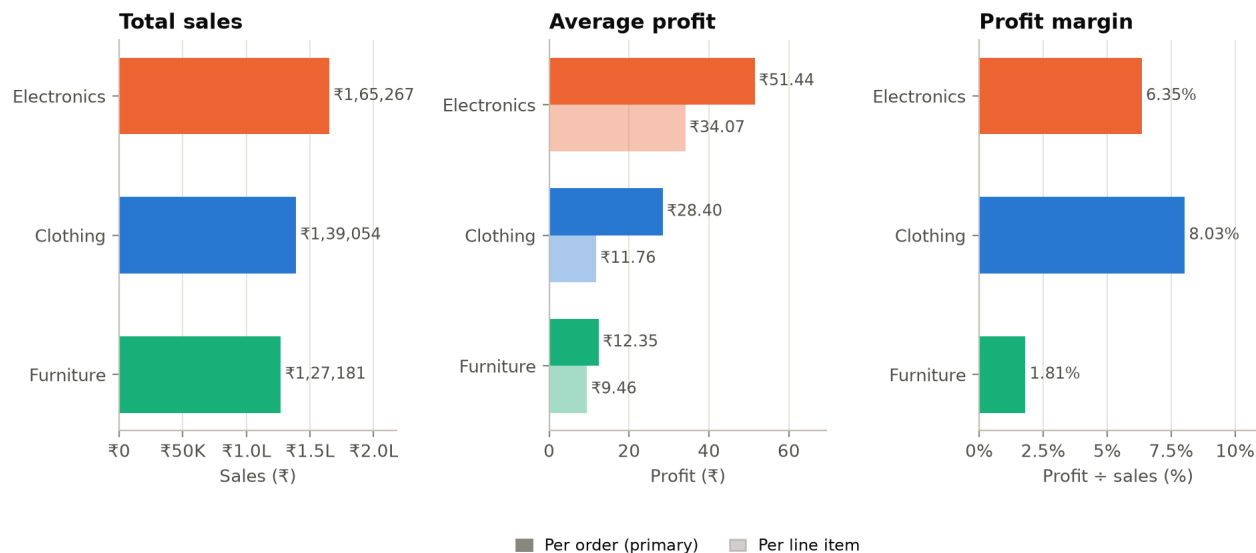


Figure 1. Category totals: sales, average profit (per order and per line) and margin.

Category	Sales	Share of sales	Profit	Share of profit	Profit / order	Profit / line	Margin
Electronics	₹1,65,267	38.3%	₹10,494	43.8%	₹51.44	₹34.07	6.35%
Clothing	₹1,39,054	32.2%	₹11,163	46.6%	₹28.40	₹11.76	8.03%
Furniture	₹1,27,181	29.5%	₹2,298	9.6%	₹12.35	₹9.46	1.81%
Total	₹4,31,502	100%	₹23,955	100%			5.55%

Verdict.

- **Electronics is the top performer on scale and order economics.** It ranks #1 on sales and on profit per order.
- **Clothing is the top performer on efficiency.** It ranks #1 on total profit and on margin.
- **Furniture underperforms on every metric.** It is last on all four.

Clothing's per-line profit (₹11.76) understates its value. Its lines are small (average unit price ₹39.55 vs ₹143.21 for Electronics), but it appears in 393 of 500 orders. That makes it the most frequent basket component and the steadiest profit pool.

2.2 Why the categories differ

1. Each category is held back by one or two sub-categories.

Two sub-categories lose money: Tables (-₹4,011) and Electronic Games (-₹1,236)

Total profit by sub-category; bars sorted within each category

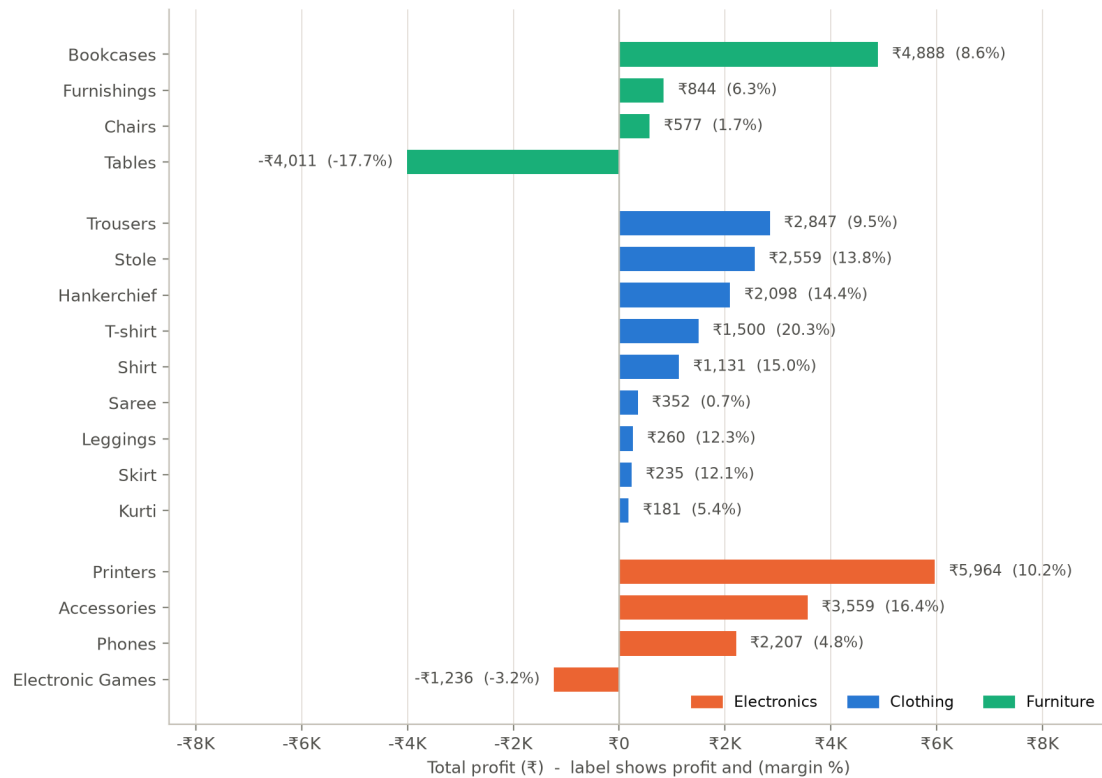


Figure 2. Profit and margin by sub-category.

- Furniture: Tables destroy the category.** Tables lose -₹4,011 on ₹22,614 of sales (-17.7% margin), with 64.7% of Tables lines in loss. Chairs barely break even (1.7% margin; 52.7% of lines in loss). Bookcases alone earn ₹4,888, which is 213% of Furniture's net profit. Without Tables, Furniture's margin would be 6.03%, close to Electronics (6.35%).
- Electronics: Electronic Games and Phones dilute a strong core.** Electronic Games lose -₹1,236 (-3.2%), and Phones run a 4.8% margin with 49.4% of lines in loss. Printers (₹5,964, 56.8% of category profit) and Accessories (16.4% margin) carry the category. Without Electronic Games, the margin would be 9.30%.
- Clothing: Sarees are the volume trap.** Sarees are 38.5% of Clothing sales but earn a 0.66% margin, with 46.7% of lines in loss. The rest of Clothing earns 12.6%, led by T-shirts (20.3%), Shirts (15.0%) and Handkerchiefs (14.4%).

2. Losses are concentrated, and in Furniture they are severe.

Furniture's loss-making lines erase 87% of what its profitable lines earn

Line-level gains and losses by category

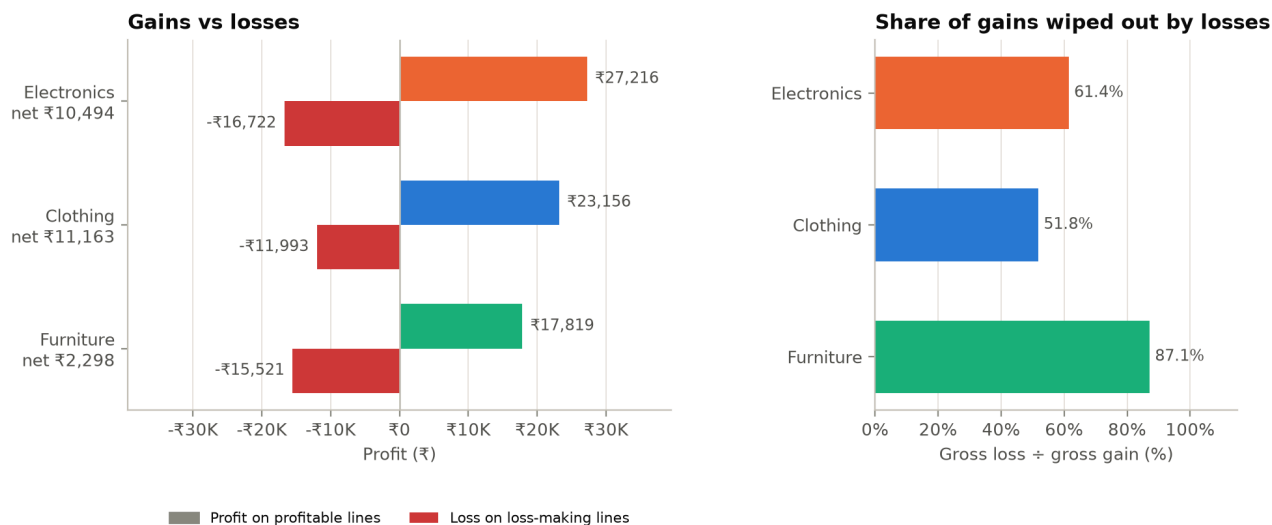


Figure 3. Gross gains from profitable lines vs gross losses from loss-making lines.

Category	Lines in loss	Loss ÷ gain	Worst 10 lines' share of losses	Lines losing more than their revenue
Electronics	39.6%	61.4%	41.5%	3
Clothing	29.4%	51.8%	28.7%	1
Furniture	42.0%	87.1%	57.6%	10

Furniture's profitable lines earn ₹17,819, and its loss-making lines give back ₹15,521 of that. Its worst 10 lines account for 57.6% of all Furniture losses. A small set of deals, not the whole range, separates Furniture from the other categories.

3. Loss lines are priced, not sized, into loss (discount proxy).

In 16 of 17 sub-categories, loss lines sell at a lower unit price (median 0.55x)

Discount proxy: the dataset has no discount field, so realised unit price (Amount ÷ Quantity) is compared

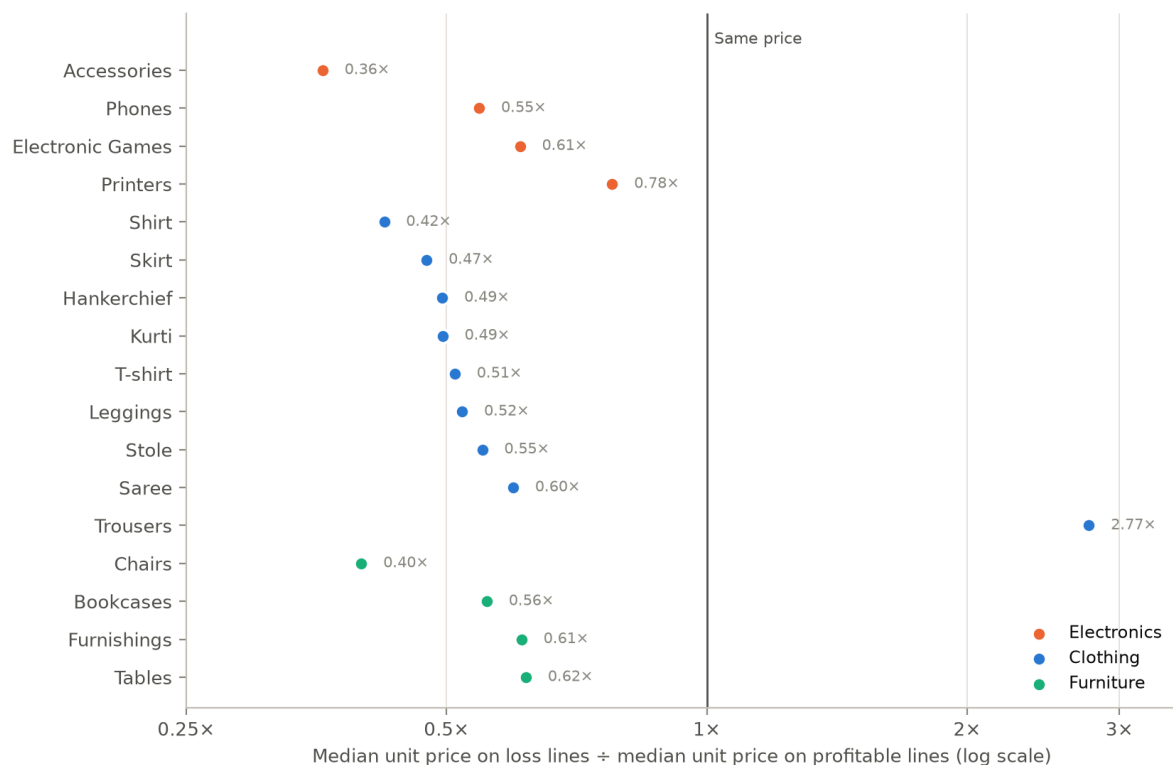


Figure 4. Median unit price on loss-making vs profitable lines within each sub-category.

In 16 of 17 sub-categories, loss-making lines sell at a lower median unit price than profitable lines (median ratio 0.55x; Chairs 0.40x, Accessories 0.36x). The one exception is Trousers (2.77x), where losses sit on high-ticket lines, which points to a cost problem rather than a discount. Median quantity is identical (3 units on loss and on profitable lines), so volume does not explain the losses. Discounting or low realised prices do.

4. Category economics differ. Furniture has a high unit price (₹134.58) but thin profit per unit (₹2.43 vs ₹9.09 for Electronics). Each discount on a big-ticket item wipes out more margin.

Actions.

1. Put a price floor on Tables and Chairs, the two Furniture sub-categories that drive the losses (Tables - ₹4,011).
2. Review Electronic Games pricing or delist it.
3. Cap Saree discounting.
4. Protect and promote Printers, Bookcases and the high-margin Clothing accessories.

3. Q1 Part 2 — Target achievement (Furniture)

3.1 Month-over-month change in target

The Furniture target ramps smoothly: every MoM change is between 0.87% and 1.89%

Red = flagged by the z-score rule ($|\text{MoM \%} - \text{mean}| > 1.5 \text{ SD}$)

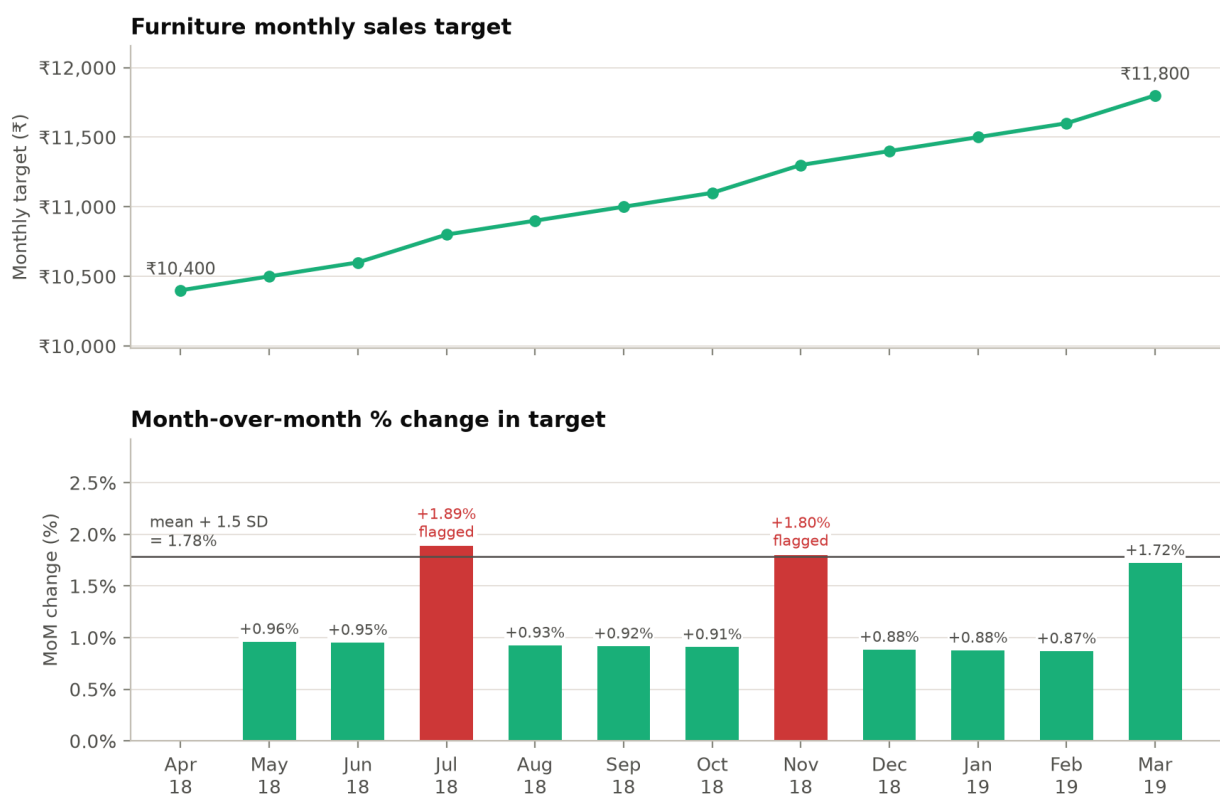


Figure 5. Furniture monthly target and its month-over-month % change.

The target rises from ₹10,400 (April 2018) to ₹11,800 (March 2019), a 13.5% increase across the year. Monthly steps are ₹100, except the ₹200 steps in Jul 2018, Nov 2018 and Mar 2019. Month-over-month change ranges from 0.87% to 1.89% (mean 1.16%, standard deviation 0.42 pp).

Fluctuation rule. A month is flagged when its MoM % is more than 1.5 standard deviations from the series mean (z-score rule; threshold 1.78%). As a materiality cross-check, a month is also flagged if $|\text{MoM}| \geq 5\%$.

Month	Target	MoM %	z-score	Flag (z-rule)
Jul 2018	₹10,800	+1.89%	1.74	Flagged
Nov 2018	₹11,300	+1.80%	1.54	Flagged
Mar 2019	₹11,800	+1.72%	1.36	Near miss
All other months		+0.87% to +0.96%	≤ -0.46	–

The z-rule flags **Jul 2018 and Nov 2018** as the significant months, with Mar 2019 just under the bar. The materiality rule flags **0 months**. The target's own fluctuations are statistically visible but commercially trivial. The planning problem is not volatility in the target; it is a target that is *too smooth* for a volatile business.

3.2 Target vs actual: where expectations and reality diverge

Flat targets miss a back-loaded year: 7 straight misses, then 4 of the last 5 months above target

Actual = sum of Furniture line Amounts per repaired order month

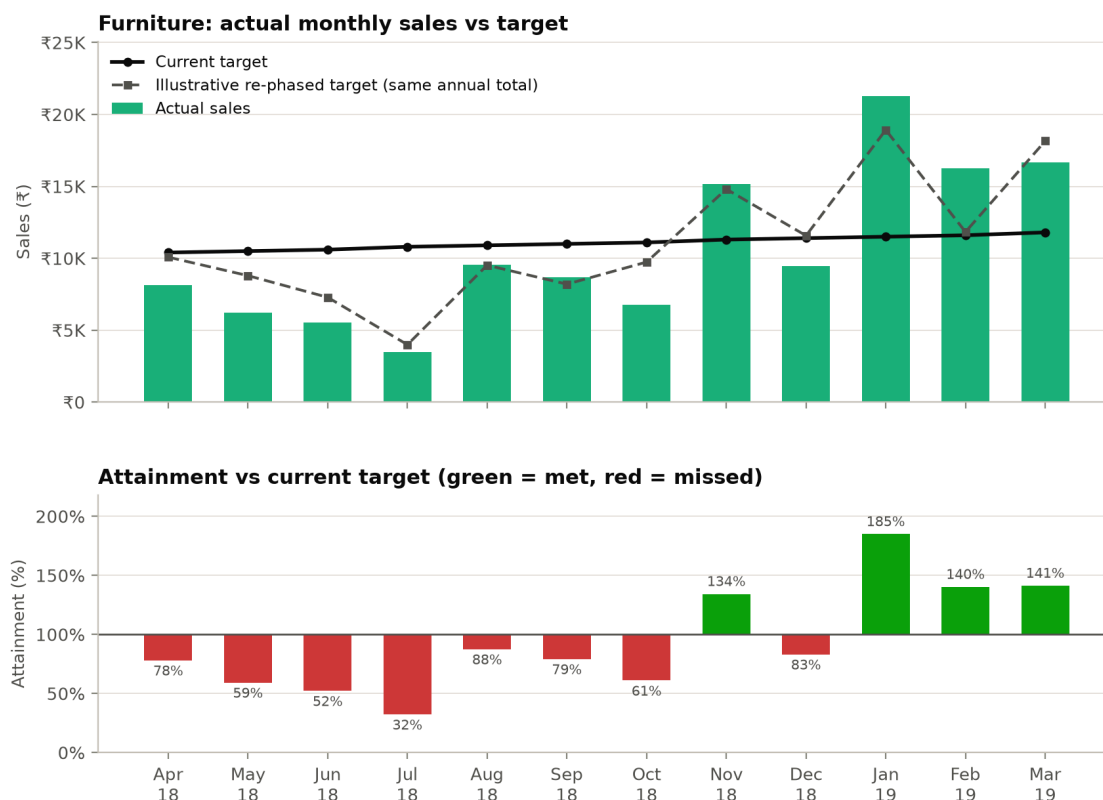


Figure 6. Actual Furniture sales vs current and illustrative re-phased target, and monthly attainment.

- **Annually, the target is about right:** ₹1,27,181 achieved against ₹1,32,900 (95.7%; gap -₹5,719).
- **Monthly, it is phased wrong.** Furniture met target in only 4 of 12 months. It missed for 7 straight months to Oct 2018 (64.2% cumulative attainment, -₹26,936 short). It then beat target in 4 of the last 5 months (136.8%, ₹21,217 over). Attainment ranges from 32% (Jul 2018) to 185% (Jan 2019).
- **The first half carries too much of the target.** The current plan puts 48.3% of the annual target into April–September, but only 32.7% of actual sales fell there. H1 attainment was 64.8%; H2 was 124.6%.
- **Actual demand is 12.5× more volatile than the target.** The coefficient of variation is 51.3% for actuals vs 4.1% for the target. Actual MoM swings run from -37.5% to +173.8% (Aug 2018).
- **A revenue-only target hides loss-making months.** Furniture lost money in 6 months (Apr 2018, May 2018, Jun 2018, Jul 2018, Sep 2018 and Oct 2018), totalling -₹9,316 on ₹38,826 of sales. The worst month was Apr 2018 at a -42.2% margin.

3.3 Aligning targets with performance

1. **Phase the annual target by seasonality, not as a straight line.** Spreading the same ₹1,32,900 across months by each month's share of *company-wide* sales gives a target from ₹3,993 (Jul 2018) to ₹18,923 (Jan 2019). Company-wide sales are used, rather than Furniture's own, to avoid fitting the target to the series it is judged against. MAPE falls from 54.4% to 19.5%, and months within $\pm 15\%$ of target rise from 1 to 6. *Caveat:* seasonality here comes from the same year (in-sample), so this is an upper bound on the gain. In practice, phase on prior-year or rolling three-year shares.
2. **Pair the revenue target with a margin guardrail.** 6 months were loss-making, so a Furniture target should also carry a floor such as "margin $\geq 0\%$ each month." Discount approval on Tables and Chairs

should be tied to that floor (Section 2).

3. **Use a target band and re-forecast quarterly.** With actual CV at 51%, a point target will be missed almost every month. A $\pm 15\%$ band, re-forecast each quarter on the latest run-rate, keeps the team focused on real deviations.
4. **Re-balance targets across categories.** Electronics finished at 128.1% of its annual target, so it was under-targeted. Clothing finished at 79.9%, so it was over-targeted. Furniture was close at 95.7%. The total ambition can stay, but the split should follow demonstrated run-rates.

4. Q1 Part 3 — Regional performance

4.1 Top 5 states by order count

Two states hold 38.2% of orders; Punjab makes the top 5 yet loses money

Top 5 states by distinct Order ID count. Red = negative

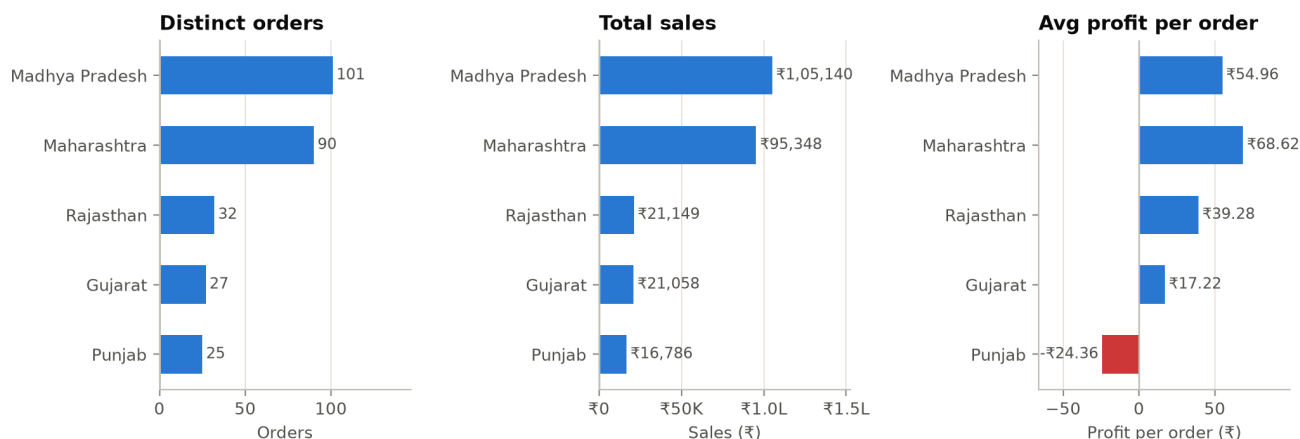


Figure 7. Top 5 states by distinct orders: order count, total sales and average profit per order.

Rank	State	Distinct orders	Total sales	Total profit	Avg profit / order	Avg profit / line	Margin
1	Madhya Pradesh	101	₹1,05,140	₹5,551	₹54.96	₹16.33	5.28%
2	Maharashtra	90	₹95,348	₹6,176	₹68.62	₹21.30	6.48%
3	Rajasthan	32	₹21,149	₹1,257	₹39.28	₹16.99	5.94%
4	Gujarat	27	₹21,058	₹465	₹17.22	₹5.34	2.21%
5	Punjab	25	₹16,786	-₹609	-₹24.36	-₹10.15	-3.63%

Average profit is per order (state profit ÷ distinct orders). Per-line is shown for reference. There is no tie at the cut-off: the next states (Delhi, Uttar Pradesh and West Bengal) have 22 orders each.

The top 5 account for 55.0% of orders and 60.1% of sales, but only 53.6% of profit. Volume is concentrated in states that earn below-average margins. Madhya Pradesh and Maharashtra alone hold 38.2% of orders and 46.5% of sales.

4.2 Regional disparities

4 states lose money; 4 smaller states earn well above their sales share

Blue = state captures more profit than its sales share; red = less

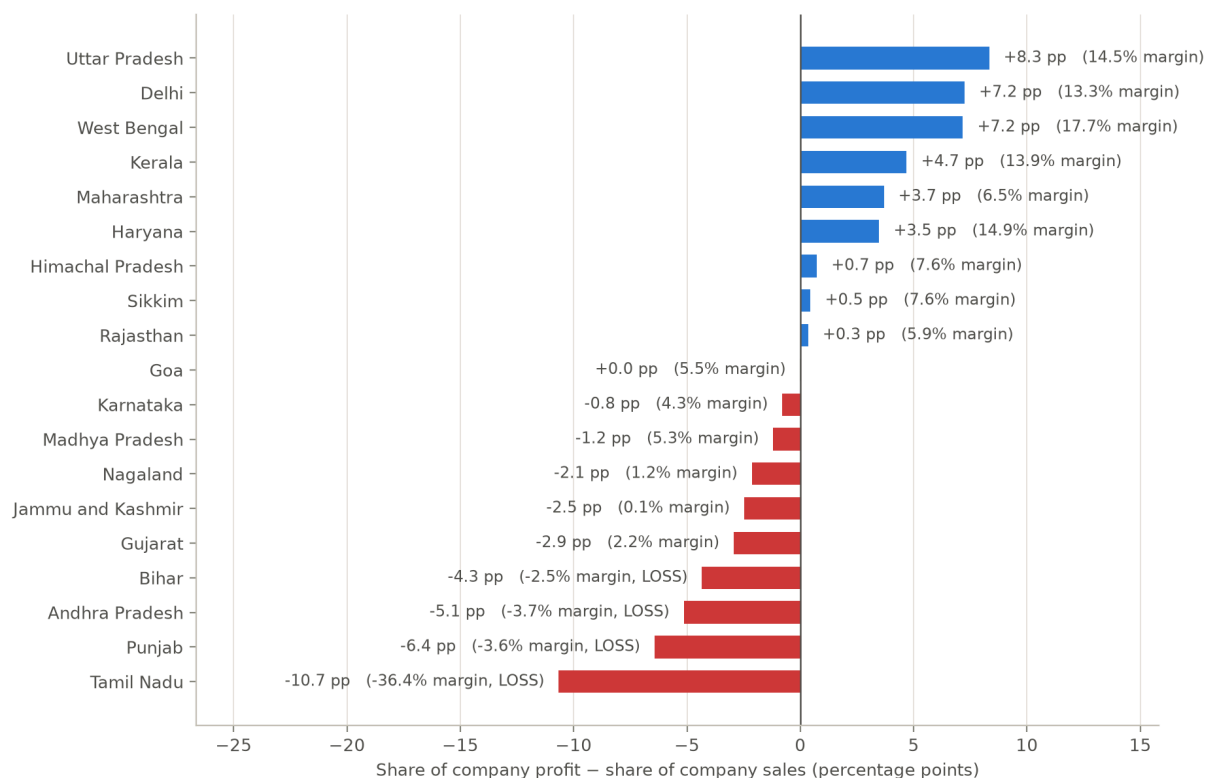


Figure 8. Each state's share of company profit minus its share of company sales.

- **Sales-to-profit mismatch.** Madhya Pradesh is the largest state (24.4% of sales) but takes a smaller share of profit (23.2%). Uttar Pradesh, Delhi, West Bengal and Kerala together earn 44.2% of company profit on 16.8% of sales (14.6% combined margin, against 5.55% for the company).
- **4 loss-making states, each sunk by a single category.**
- Tamil Nadu: -₹2,216, -36.4% margin, driven by Furniture (-₹2,487).
- Punjab: -₹609, driven by Electronics (-₹2,132, -34.8%).
- Andhra Pradesh: -₹496, driven by Furniture (-₹1,568).
- Bihar: -₹321, driven by Electronics (-₹814).
- **Losses are concentrated at city level inside the top states.** In Gujarat, Punjab and Rajasthan, one city makes money and the other loses it (Figure 9):

State	Loss-making city	Sales	Profit	Margin	Profitable city	Margin
Gujarat	Ahmedabad	₹14,230	-₹880	-6.2%	Surat	19.7%
Punjab	Chandigarh	₹12,279	-₹1,153	-9.4%	Amritsar	12.1%
Rajasthan	Jaipur	₹10,076	-₹753	-7.5%	Udaipur	18.2%

- **The two largest cities are thin-margin.** Mumbai (₹61,867, 2.65%) and Indore (₹79,069, 5.26%) bring in 32.7% of company sales but only 24.2% of profit.

Mumbai and Indore carry the volume at thin margins; Chennai is the deepest loss

Each bubble is a city (size = distinct orders). Labels show priority cities and their margin

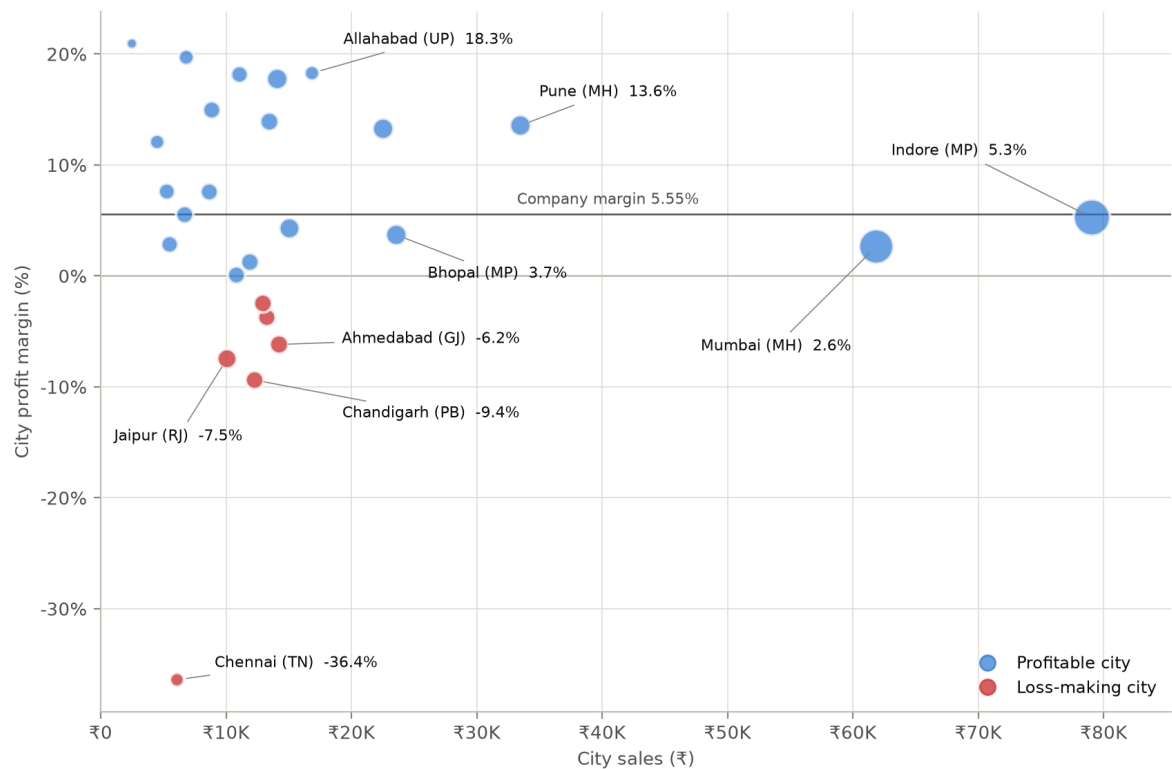


Figure 9. City sales vs margin; bubble size is distinct orders.

4.3 Where to focus

Priority	Where	Why (number)	Lever
1. Margin repair at scale	Mumbai	₹61,867 sales at 2.65% margin; 41.1% of lines in loss vs 28.9% in Pune	Matching Pune's margin adds about ₹6,750; matching the company average adds ₹1,798
2. Stop the bleed	Chennai (Tamil Nadu)	-₹2,216 on ₹6,087 (-36.4%)	Suspend loss-making Furniture lines (-₹2,487) until pricing or logistics are fixed
3. Fix the loss cities inside the top 5	Ahmedabad, Chandigarh and Jaipur	Combined -₹2,786, while their sister cities earn double-digit margins	Copy the sister city's category mix and discount policy; for Punjab, start with Electronics
4. Defend the volume base	Indore and Bhopal	Indore is the largest city (75.2% of Madhya Pradesh); Bhopal margin 3.7%	Keep Indore near company margin; lift Bhopal
5. Grow where profit is easy	Allahabad, Delhi, Kolkata, Thiruvananthapuram	Uttar Pradesh, Delhi, West Bengal and Kerala: 14.6% combined margin on only 82 orders	Acquisition spend: highest return per new order

Data note: 3 orders list city "Delhi" under state Madhya Pradesh. They were kept as recorded and do not change any ranking. Chandigarh appears under both Punjab (16 orders) and Haryana (14); the two are treated separately.

5. Q2 — App exploration

How to read this section. This is a *pre-session framework* built only from Jar's public material (website, store listing, press). Every observation tagged **[VERIFY]** must be confirmed, corrected or replaced from my own hands-on session before submission, and each has a screenshot slot. The checklist is in Appendix D.

Context from public sources. Jar reports more than 35 million registered users across 12,000 PIN codes. About 60% are in tier-2 and tier-3 towns, and more than 95% are first-time formal savers (TechCrunch, Sept 2025). Jar says the app supports nine Indian languages. Its site lists daily (from ₹10), weekly (from ₹50) and monthly (from ₹100) automated saving, Brink's as vault partner, ICICI Lombard insurance and Vistra as independent auditor/trustee.

5.1 Five things that work well

#	What works	Why it works (UX principle)	Evidence to capture
1	₹10 daily saving on UPI AutoPay. A one-time mandate replaces a daily decision.	<i>Habit loop and default effect.</i> Saving becomes the default, so the cue, routine and reward run without willpower. ₹10 removes affordability friction for first-time savers.	[VERIFY] Taps from install to first AutoPay mandate; is ₹10 the pre-selected amount?
2	Trust signals placed where money moves. Vault partner, insurer, trustee audit, 99.9% purity.	<i>Trust signals in fintech.</i> Named third parties stand in for regulation the user cannot see; showing them at the moment of payment reduces perceived risk.	[VERIFY] Are Brink's / ICICI Lombard / Vistra shown on the payment screen or only on the website?
3	Vernacular, low-literacy-friendly interface.	<i>Onboarding friction / accessibility.</i> Native language and large numerals cut cognitive load for the tier-2/3 majority.	[VERIFY] Language picker at first launch? Is the whole flow translated, including KYC and errors?
4	Visible, reversible exit: sell anytime, or convert to coins or Nek jewellery.	<i>Loss aversion and reversibility.</i> Users commit more when undoing is easy and visible; delivery makes "digital" gold tangible.	[VERIFY] Taps to sell; time for money to reach the bank; where delivery options appear.
5	Round-ups and nudges that save spare change from everyday spends.	<i>Mental accounting / pain of paying.</i> Small amounts attached to spends that already happened feel costless; streaks and nudges reinforce the loop.	[VERIFY] Is round-up still offered? Is it SMS-based or native to Jar UPI (@jarunity)? Which streak or reward mechanics appear?

Screenshot slots 5.1-1 to 5.1-5: home screen with daily-saving card · payment screen with trust badges · language picker · sell/withdraw screen · round-up or streak screen

5.2 Five areas to improve

#	Improvement	Reasoning (UX principle)	What to check / propose
1	Show the all-in cost before purchase: 3% GST plus the buy-sell spread, as "if you sold right now you'd get ₹X."	<i>Information scent / honest pricing.</i> Third-party reviews report spreads of roughly 2.5–6% across digital gold apps. Discovering the cost on exit breaks trust. RBI and industry voices have called for standardised disclosure.	[VERIFY] Are buy price, sell price and GST shown together before payment?

#	Improvement	Reasoning (UX principle)	What to check / propose
2	A plain-language "what protects my gold" page.	<i>Trust repair after a public shock.</i> SEBI's Nov 2025 advisory said digital gold is outside its oversight. Users need a calm explainer: custody, audit certificate date, insurance, what happens if Jar fails.	[VERIFY] Is there an in-app explainer? Is it findable in two taps from the gold balance?
3	Withdrawal status and error recovery: a progress tracker with an ETA, and KYC prompted <i>before</i> the first withdrawal.	<i>Visibility of system status; help users recover from errors</i> (Nielsen). Public reviews cite slow withdrawals and surprise document requests at exit, which is the worst moment for friction.	[VERIFY] When is KYC asked? Is there a tracker and a support path inside the flow?
4	Pause, skip or lower the daily saving without cancelling the mandate.	<i>Habit loops need slack.</i> A user in a cash crunch who can only cancel churns for good; a seven-day pause preserves the habit and the mandate.	[VERIFY] Options shown when a user tries to stop AutoPay.
5	Goals with progress (Diwali, wedding, Akshaya Tritiya).	<i>Goal-gradient and endowed-progress effects.</i> Savings tied to a named goal and a progress bar keep going longer than an open-ended balance.	[VERIFY] Do goals exist? If yes, replace this with the weakest flow observed in the session.

Screenshot slots 5.2-1 to 5.2-5: pre-payment price breakdown · safety/FAQ page · withdrawal flow · AutoPay management · goals (or absence)

[VERIFY] Replace-if-wrong rule. If any "works well" item is broken in practice, move it to 5.2 with what you observed. Then promote the strongest item from your session notes (e.g., gamified rewards, gold-price alerts) into 5.1.

6. Q3 — Product exploration

6.1 Starting point: Jar's real strengths, and one correction

Jar's moat is behavioural, not product. It has (i) **automation**: UPI AutoPay mandates on a base of more than 35 million registered users; (ii) **micro-ticket access** at ₹10; (iii) **UX simplicity** in nine languages for a 60% tier-2/3 base; and (iv) **trust earned from first-time savers**, backed by named custodians.

The brief describes this trust as earned in a *regulated* category. That needs a correction. SEBI's advisory of 8 November 2025 states that digital gold is neither a security nor a regulated commodity derivative, so it sits outside SEBI's oversight. In 2026, Jar's gold business has also faced a state police investigation and related court proceedings over frozen accounts. The matter is unresolved, and no finding has been made against the company (Inc42). The strategic implication is clear: **Jar's next growth should move its automation and trust onto regulated rails**. That defends the core and opens new revenue.

Start with regulated gold and goal-linked Nek savings; gate credit behind compliance

Analyst-judgement scores (1-5). Priority = impact ÷ effort

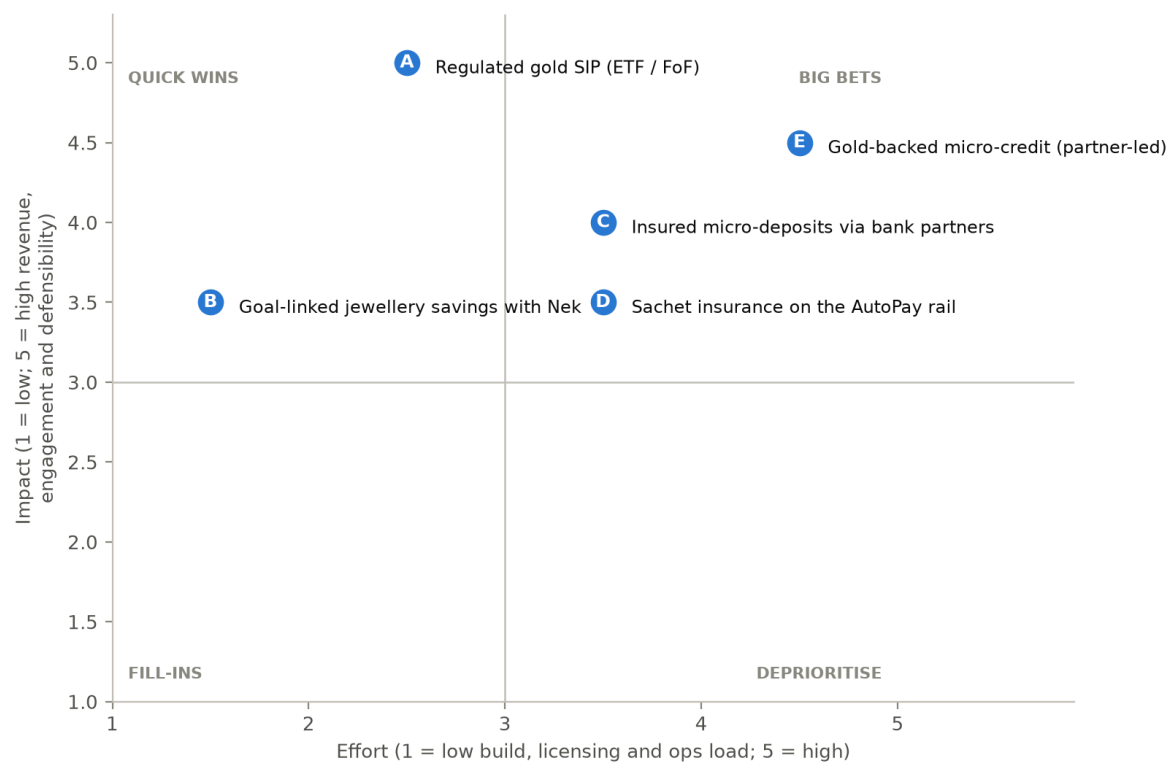


Figure 10. Effort vs impact for the five opportunities (analyst-judgement scores, 1–5).

Rank	Opportunity	Impact	Effort	Quadrant	Horizon
1	B. Goal-linked jewellery savings with Nek	3.5	1.5	Quick win / do now	Now (0-3 months)
2	A. Regulated gold SIP (ETF / FoF)	5.0	2.5	Quick win / do now	Now (0-6 months)
3	C. Insured micro-deposits via bank partners	4.0	3.5	Big bet / sequence	Next (6-12 months)
4	D. Sachet insurance on the AutoPay rail	3.5	3.5	Big bet / sequence	Next (9-15 months)
5	E. Gold-backed micro-credit (partner-led)	4.5	4.5	Big bet / sequence	Later (gated, 12+ months)

6.2 Opportunity cards

A. Regulated gold SIP (Gold ETF / Gold fund-of-funds) — the strategic priority

- **User problem.** Savers who now know digital gold is unregulated want the same "small daily gold" habit with investor protection, and they don't know how to buy an ETF.
- **Why Jar wins.** It already holds the AutoPay mandate and the gold-saving intent. A "Gold+ (SEBI-regulated)" toggle moves an existing habit instead of acquiring a new customer. Low-ticket mutual-fund SIPs (₹100–500 at many AMCs) fit Jar's micro-ticket DNA.
- **Integration path.** Register as a mutual-fund distributor, or partner with a registered execution platform → offer the gold FoF/ETF as a second "jar" on the same home card → let users route new daily savings or migrate balances, with a cost comparison shown.
- **Monetisation.** Distribution trail on regular plans, or a flat platform fee on direct plans. Lower per-rupee take than the digital-gold spread, but defensible and recurring.
- **Key risk.** Cannibalising the higher-margin digital-gold spread. Distributor conduct rules (suitability, commission disclosure).
- **Success metric.** Share of active savers with a regulated gold SIP within 6 months; regulated share of total gold AUM.

B. Goal-linked jewellery savings with Nek — the quick win

- **User problem.** Indian households save for gold jewellery around weddings, Akshaya Tritiya and Dhanteras, often through jeweller 11-month schemes that lock money to one store.
- **Why Jar wins.** Nek is already Jar's own jewellery brand. Automation plus a named goal ("Wedding necklace – ₹45,000 by Feb") turns open-ended saving into a purchase the user can see coming.
- **Integration path.** Goal creation on the home card → progress bar with a projected date → one-tap redemption into Nek at checkout, with a making-charge discount for completed goals.
- **Monetisation.** Nek jewellery margin and making charges. Lower CAC for Nek, because the buyer is pre-funded and pre-committed.
- **Key risk.** Must be framed as accumulation of the user's own gold, not a deposit scheme, so that it stays clear of deposit and unregulated-scheme rules. Fulfilment quality at Nek.
- **Success metric.** Goal completion rate; share of completed goals redeemed at Nek; 12-month saver retention for goal vs non-goal users.

C. Insured micro-deposits via bank partners

- **User problem.** First-time savers want a safe, fixed-return option for emergency money. Bank FDs feel out of reach for them.
- **Why Jar wins.** The same mandate rail sweeps daily ₹10s into a recurring deposit or FD at partner small finance banks. Jar already has a banking relationship through its UPI launch with Unity Small Finance Bank. Deposits are insured by DICGC up to ₹5 lakh per depositor per bank, which is a regulated trust anchor Jar can explain simply.
- **Integration path.** Bank-partner FD/RD marketplace inside the app → "safe jar" next to the gold jar → auto-split rules (e.g., 70% gold, 30% deposit).
- **Monetisation.** Distribution fee from partner banks on deposits sourced; cross-sell lift.
- **Key risk.** Bank-partner concentration and outsourcing/KYC obligations. Rates must be displayed transparently, with no "returns" language bleeding into the unregulated gold product.
- **Success metric.** Deposit balances per active saver; share of savers holding both gold and a deposit; early-withdrawal rate.

D. Sachet insurance on the AutoPay rail

- **User problem.** Tier-2/3 and gig workers are under-insured. Annual premiums feel large, while ₹2–5 a day does not.
- **Why Jar wins.** Daily auto-debit is exactly how sachet premiums must be collected. Jar's tone of voice is suited to explaining cover in plain language.
- **Integration path.** IRDAI-registered corporate-agent licence or broker partner → add-on toggles (personal accident, hospital cash) at the moment a saving streak is set up.
- **Monetisation.** Commission within IRDAI's expense limits.
- **Key risk.** Mis-selling and claim-rejection stories would damage the core trust asset. Bundling rules.
- **Success metric.** Attach rate; claim settlement turnaround; persistency at 13 months.

E. Gold-backed micro-credit (partner-led) — gated

- **User problem.** Small emergencies are funded by informal lenders or by selling gold at a bad moment.
- **Why Jar wins.** The collateral is already vaulted and valued daily, so a small loan against holdings needs no branch visit or assayer. That beats the physical gold-loan journey on speed.
- **Integration path.** Jar as lending service provider to an RBI-regulated bank or NBFC → "borrow instead of sell" offered on the sell screen.
- **Monetisation.** Sourcing fee or revenue share on disbursals.
- **Key risk.** The highest regulatory load of the five: digital-lending and gold-loan rules, and the open legal question of pledging *digital* gold that the RBI does not regulate. It should start only after the regulated rails (A, C) are live and the current proceedings are resolved.
- **Success metric.** Loans per 1,000 active savers, 90+ day delinquency, and share of borrowers who keep saving.

Considered and deprioritised. Digital silver (same regulatory exposure as gold, little new value). B2B payroll-linked saving for gig platforms (high CAC payoff but long enterprise sales cycles; revisit after A–C).

6.3 Sequenced roadmap

Horizon	Moves	Milestone
0–6 months	B (Nek goals) and A (regulated gold SIP)	Regulated product live; the first goal cohorts complete
6–15 months	C (insured deposits), then D (sachet insurance)	Majority of new savings on regulated or insured rails
12+ months, gated	E (gold-backed credit) via a regulated partner	Credit launched with clean compliance and delinquency within plan

6.4 Sources (Sections 5–6)

- TechCrunch, "Indian fintech Jar turns profitable by helping millions save in gold", 18 Sep 2025.
- Jar website (myjar.app): saving modes and minimums, custody/insurance/trustee partners. Accessed Oct 2026.
- Groww; Inc42, "India's Digital Gold Rush Gets Regulatory Reality Check", 12 Nov 2025: SEBI advisory of 8 Nov 2025.
- Inc42, "Karnataka HC quashes sessions court orders to defreeze Jar's accounts", 2026: investigation and court proceedings.
- Entracker: Jar enters UPI payments through BharatPe and Unity Small Finance Bank, 2025.

- Inc42 / Entrackr: FY25 financials; Nek jewellery revenue.
- Third-party reviews and spread comparisons (IPOMarket, Jupiter, Analytics Insight): directional only.

Appendix A — Assumptions log

#	Assumption	Why	Effect if wrong
A1	Order Date datetime cells have day and month swapped; text cells are dd-mm-yyyy	Only reading that makes dates monotonic in sequential Order ID	Part 2 monthly actuals would shift months
A2	Sales Target "day" component encodes the year (18 → 2018)	Months then run Apr-2018 to Mar-2019, exactly the order period	No overlap for target-vs-actual
A3	Target values are monthly <i>sales</i> (Amount) targets in ₹	Same order of magnitude as monthly category sales	Attainment metrics would not apply
A4	"Average profit per order" = profit ÷ distinct orders containing the group	Matches the brief's wording; per-line reported alongside	Magnitudes change; rankings do not
A5	Realised unit price is a valid proxy for discounting	No discount column exists	The discount narrative would weaken; loss concentration still holds
A6	3 "Delhi, Madhya Pradesh" orders kept as recorded	Brief analyses the State field	Madhya Pradesh loses 3 orders; still #1
A7	Re-phased target uses same-year company seasonality	Only one year of data	MAPE improvement is an upper bound
A8	Q3 impact/effort scores are analyst judgement	No internal Jar data	Ranking is a starting point for discussion

Appendix B — Detail tables

B1. Sub-category performance

Category	Sub-category	Sales	Profit	Margin	Lines	Orders	Loss lines	Profit / line
Clothing	Trousers	₹30,039	₹2,847	9.5%	39	37	38.5%	₹73.00
Clothing	Stole	₹18,546	₹2,559	13.8%	192	157	27.1%	₹13.33
Clothing	Hankerchief	₹14,608	₹2,098	14.4%	198	138	21.2%	₹10.60
Clothing	T-shirt	₹7,382	₹1,500	20.3%	77	70	15.6%	₹19.48
Clothing	Shirt	₹7,555	₹1,131	15.0%	69	66	17.4%	₹16.39
Clothing	Saree	₹53,511	₹352	0.7%	210	156	46.7%	₹1.68
Clothing	Leggings	₹2,106	₹260	12.3%	53	49	30.2%	₹4.91
Clothing	Skirt	₹1,946	₹235	12.1%	64	56	28.1%	₹3.67
Clothing	Kurti	₹3,361	₹181	5.4%	47	41	29.8%	₹3.85
Electronics	Printers	₹58,252	₹5,964	10.2%	74	67	33.8%	₹80.59
Electronics	Accessories	₹21,728	₹3,559	16.4%	72	65	22.2%	₹49.43
Electronics	Phones	₹46,119	₹2,207	4.8%	83	71	49.4%	₹26.59
Electronics	Electronic Games	₹39,168	-₹1,236	-3.2%	79	73	50.6%	-₹15.65
Furniture	Bookcases	₹56,861	₹4,888	8.6%	79	72	27.8%	₹61.87
Furniture	Furnishings	₹13,484	₹844	6.3%	73	66	41.1%	₹11.56
Furniture	Chairs	₹34,222	₹577	1.7%	74	64	52.7%	₹7.80
Furniture	Tables	₹22,614	-₹4,011	-17.7%	17	16	64.7%	-₹235.94

B2. Furniture: target vs actual by month

Month	Target	MoM % (target)	Actual	Gap	Attainment	Furniture profit	Re-phased target
Apr 2018	₹10,400	—	₹8,121	-₹2,279	78.1%	-₹3,425	₹10,079
May 2018	₹10,500	+0.96%	₹6,220	-₹4,280	59.2%	-₹794	₹8,792
Jun 2018	₹10,600	+0.95%	₹5,532	-₹5,068	52.2%	-₹856	₹7,287
Jul 2018	₹10,800	+1.89%	₹3,483	-₹7,317	32.2%	-₹457	₹3,993
Aug 2018	₹10,900	+0.93%	₹9,538	-₹1,362	87.5%	₹443	₹9,517
Sep 2018	₹11,000	+0.92%	₹8,704	-₹2,296	79.1%	-₹2,468	₹8,201
Oct 2018	₹11,100	+0.91%	₹6,766	-₹4,334	61.0%	-₹1,316	₹9,737
Nov 2018	₹11,300	+1.80%	₹15,165	₹3,865	134.2%	₹3,945	₹14,810
Dec 2018	₹11,400	+0.88%	₹9,474	-₹1,926	83.1%	₹187	₹11,574
Jan 2019	₹11,500	+0.88%	₹21,257	₹9,757	184.8%	₹3,284	₹18,923
Feb 2019	₹11,600	+0.87%	₹16,262	₹4,662	140.2%	₹2,168	₹11,834
Mar 2019	₹11,800	+1.72%	₹16,659	₹4,859	141.2%	₹1,587	₹18,152

B3. All states

Rank	State	Orders	Sales	Profit	Margin	Profit / order	Sales share	Profit share
1	Madhya Pradesh	101	₹1,05,140	₹5,551	5.3%	₹54.96	24.4%	23.2%
2	Maharashtra	90	₹95,348	₹6,176	6.5%	₹68.62	22.1%	25.8%
3	Rajasthan	32	₹21,149	₹1,257	5.9%	₹39.28	4.9%	5.2%
4	Gujarat	27	₹21,058	₹465	2.2%	₹17.22	4.9%	1.9%
5	Punjab	25	₹16,786	-₹609	-3.6%	-₹24.36	3.9%	-2.5%
6	Delhi	22	₹22,531	₹2,987	13.3%	₹135.77	5.2%	12.5%
7	Uttar Pradesh	22	₹22,359	₹3,237	14.5%	₹147.14	5.2%	13.5%
8	West Bengal	22	₹14,086	₹2,500	17.7%	₹113.64	3.3%	10.4%
9	Karnataka	21	₹15,058	₹645	4.3%	₹30.71	3.5%	2.7%
10	Kerala	16	₹13,459	₹1,871	13.9%	₹116.94	3.1%	7.8%
11	Bihar	16	₹12,943	-₹321	-2.5%	-₹20.06	3.0%	-1.3%
12	Andhra Pradesh	15	₹13,256	-₹496	-3.7%	-₹33.07	3.1%	-2.1%
13	Nagaland	15	₹11,903	₹148	1.2%	₹9.87	2.8%	0.6%
14	Jammu and Kashmir	14	₹10,829	₹8	0.1%	₹0.57	2.5%	0.0%
15	Haryana	14	₹8,863	₹1,325	14.9%	₹94.64	2.1%	5.5%
16	Himachal Pradesh	14	₹8,666	₹656	7.6%	₹46.86	2.0%	2.7%
17	Goa	14	₹6,705	₹370	5.5%	₹26.43	1.6%	1.5%
18	Sikkim	12	₹5,276	₹401	7.6%	₹33.42	1.2%	1.7%
19	Tamil Nadu	8	₹6,087	-₹2,216	-36.4%	-₹277.00	1.4%	-9.3%

B4. Cities in the top-5 and loss-making states

State	City	Orders	Sales	Profit	Margin	Share of state sales	Loss lines
Andhra Pradesh	Hyderabad	15	₹13,256	-₹496	-3.7%	100.0%	26.2%
Bihar	Patna	16	₹12,943	-₹321	-2.5%	100.0%	38.7%
Gujarat	Ahmedabad	17	₹14,230	-₹880	-6.2%	67.6%	40.3%
Gujarat	Surat	10	₹6,828	₹1,345	19.7%	32.4%	8.0%
Madhya Pradesh	Indore	76	₹79,069	₹4,159	5.3%	75.2%	36.0%
Madhya Pradesh	Bhopal	22	₹23,583	₹871	3.7%	22.4%	39.4%
Madhya Pradesh	Delhi	3	₹2,488	₹521	20.9%	2.4%	0.0%
Maharashtra	Mumbai	68	₹61,867	₹1,637	2.6%	64.9%	41.1%
Maharashtra	Pune	22	₹33,481	₹4,539	13.6%	35.1%	28.9%
Punjab	Chandigarh	16	₹12,279	-₹1,153	-9.4%	73.2%	42.2%
Punjab	Amritsar	9	₹4,507	₹544	12.1%	26.8%	13.3%
Rajasthan	Udaipur	13	₹11,073	₹2,010	18.2%	52.4%	10.0%

State	City	Orders	Sales	Profit	Margin	Share of state sales	Loss lines
Rajasthan	Jaipur	19	₹10,076	-₹753	-7.5%	47.6%	27.3%
Tamil Nadu	Chennai	8	₹6,087	-₹2,216	-36.4%	100.0%	40.0%

Appendix C — Reproducibility

- **Run:** `python main.py`. This regenerates every table in `outputs/tables/`, every figure in `outputs/figures/`, `outputs/report_numbers.json`, this report's Markdown and HTML in `report/`, and the PDF. Every number in this report is injected from `report_numbers.json`. The build fails if any placeholder is unresolved, and a final check compares numbers in the PDF text against the computed values.
- **Environment:** Python 3.12.1; pandas 3.0.6; matplotlib 3.11.2; PDF rendered by headless Chromium.
- **Data:** raw Excel files in `data/` are read-only; all repairs happen in memory (Section 1.2).

Appendix D — Q2 hands-on checklist (≈ 20 minutes)

1. **Install → first saving (time it).** Count taps and screens to the first AutoPay mandate. Note the default amount and any language prompt. (5.1 #1, #3)
2. **Payment screen.** Screenshot the trust badges shown, if any. Is the buy price split into gold value, GST and spread? (5.1 #2, 5.2 #1)
3. **Sell ₹10 worth.** Note the sell price vs buy price, taps, KYC prompts, time for money to reach the bank, and status messages. (5.1 #4, 5.2 #3)
4. **Try to stop daily saving.** Are pause, skip or lower options offered before cancel? (5.2 #4)
5. **Search for safety information.** From the gold balance, how many taps to custody, audit and insurance details? Any mention of the SEBI advisory? (5.2 #2)
6. **Explore engagement.** Round-ups, streaks, spins and rewards, goals, notifications on day 1. (5.1 #5, 5.2 #5)
7. **Cross-sell surfaces.** Where do Nek, UPI, loans or insurance appear? Do they crowd out the saving card?
8. **Replace or confirm.** For each [VERIFY] row, mark ✓ confirmed / ✎ changed / X replaced, and drop the screenshot into its slot.